

Code:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int n, head, seek_time = 0;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);

    int request_queue[n];

    printf("Enter the disk request queue:\n");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", &request_queue[i]);
    }

    printf("Enter the initial position of the disk head: ");
    scanf("%d", &head);

    // Sort the request queue
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (request_queue[i] > request_queue[j])
            {
                int temp = request_queue[i];
                request_queue[i] = request_queue[j];
                request_queue[j] = temp;
            }
        }
    }

    // SCAN Scheduling
    printf("\nSCAN (Elevator) Disk Scheduling:\n");

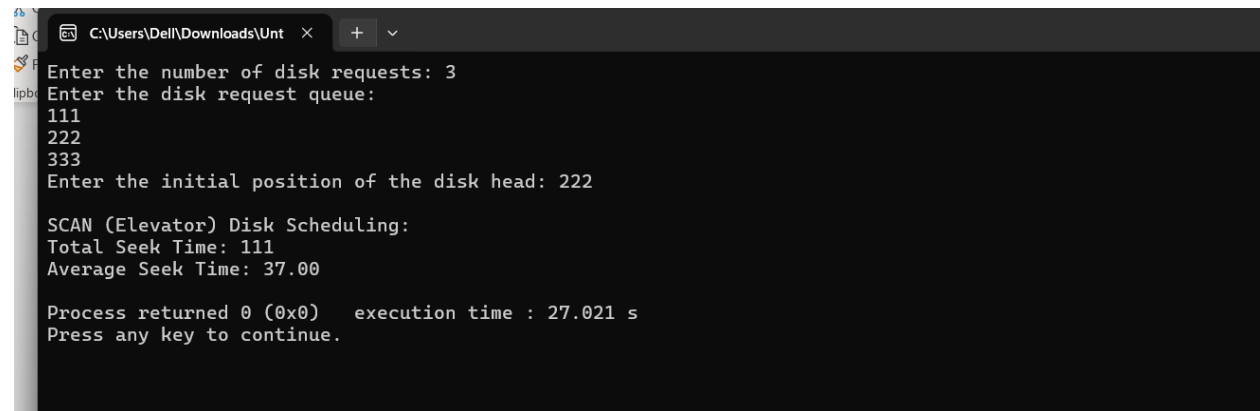
    int start = 0;
    int end = n - 1;
    int current_direction = 1; // 1 = right, -1 = left

    while (start <= end)
    {
        if (current_direction == 1)
        {
            for (int i = start; i <= end; i++)
            {
                if (request_queue[i] >= head)
                {
                    seek_time += abs(head - request_queue[i]);
                    head = request_queue[i];
                    start = i + 1;
                    break;
                }
            }
            current_direction = -1;
        }
        else
        {
            for (int i = end; i >= start; i--)
            {
                if (request_queue[i] <= head)
                {
                    seek_time += abs(head - request_queue[i]);
                    head = request_queue[i];
                    end = i - 1;
                    break;
                }
            }
            current_direction = 1;
        }
    }

    printf("Total Seek Time: %d\n", seek_time);
    printf("Average Seek Time: %.2f\n", (float)seek_time / n);

    return 0;
}
```

Output:



```
C:\Users\Dell\Downloads\Unt > Enter the number of disk requests: 3
Enter the disk request queue:
111
222
333
Enter the initial position of the disk head: 222

SCAN (Elevator) Disk Scheduling:
Total Seek Time: 111
Average Seek Time: 37.00

Process returned 0 (0x0)   execution time : 27.021 s
Press any key to continue.
```